



Modelling Financial Markets

Chapter 9: Realised Volatility



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- BV: Bipower Variation
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CI: Confidence Interval
- CLT: Central Limit Theorem
- DJIA: Dow Jones Industrial Average
- DM: Diebold–Mariano (test)
- ETF: Exchange-Traded Fund
- EWMA: Exponentially Weighted Moving Average
- FDR: False Discovery Rate
- FOMC: Federal Open Market Committee
- FRED: Federal Reserve Economic Data
- FWER: Family-Wise Error Rate
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GMM: Generalized Method of Moments
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR: Heterogeneous AutoRegressive (model)
- HAR-CJ: HAR model with Continuous and Jump components
- HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- HARQ: HAR model with Realised Quarticity
- HEAVY: High-frEQUENCY-bAsed VolatilitY (models)
- IQ: Integrated Quarticity
- LLM: Large Language Model
- MAE: Mean Absolute Error

Acronyms used in this material (2/2)

- ▣ MCS: Model Confidence Set
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ RF: Random Forest
- ▣ RK: Realised Kernel
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ SHAR: Semivariance HAR model
- ▣ TQ: Tripower Quarticity
- ▣ TRV: Truncated Realised Variance
- ▣ TSRV: Two-Scale Realised Variance
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility IndeX
- ▣ VRP: Variance Risk Premium

Chapter Route and Outcomes

Route of this chapter

- Part I, **measurement and diagnosis**: realised variance and its precision, microstructure noise, jumps
- Part II, **modelling and forecasting**: stylised facts, the HAR model, forecast evaluation against GARCH, rough volatility, a machine-learning case study, one pass of the AI-assisted research loop
- Asymptotic theory, noise- and jump-robust estimators and model extensions sit in the 
▶ **Appendix** at the end

After this chapter you can

- compute daily realised variance, its standard error and a confidence interval from intraday prices
- diagnose microstructure noise with a signature plot and test for jumps with FDR control
- estimate log-HAR and compare variance forecasts with QLIKE, Diebold–Mariano, Mincer–Zarnowitz and the model confidence set

Volatility Is Latent: Can We Measure It?



Torben G. Andersen, 2012

Photo: Soralum (2012); CC BY-SA 4.0;

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- Volatility drives risk (Chapters 7–8), option prices (Chapter 12) and portfolio weights (Chapter 4), but it is never observed
 - ▶ One daily return is a single draw from a distribution whose dispersion we want to estimate
 - ▶ GARCH (Chapter 5) **infers** today's variance from past daily returns through a model
- Idea: inside one day there are hundreds of price changes; the sum of their squares **measures** the day's variance almost without a model Andersen & Bollerslev (1998)
 - ▶ Early idea: Merton (1980) showed that variance, unlike the mean, is estimated better with more frequent data
- Two different objects: realised variance is an **ex-post measurement** of day t , known after the close; a GARCH variance is a **conditional forecast** of day t made with the information of day $t - 1$
- Measure first, then model: Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001) study the distribution of realised volatility; Andersen, Bollerslev, Diebold & Labys (2003) forecast it directly; textbook treatment: Franke, Härdle & Hafner (2019)

Data Used in This Chapter

Series	Frequency	Period	Days
SPY (ETF on the S&P 500)	5-minute bars, 09:30–16:00 New York time	2020–2026	1 489
Bitcoin	5-minute bars, 24-hour UTC days	2024–2026	659
SPY, Bitcoin	daily closes	2000/2014–2026	for GARCH
VIX	daily close	2020–2026	implied volatility

- Up to 18 September 2026; returns $r_i = 100(\ln P_i - \ln P_{i-1})$ in %, P_i : the price at the end of bar i ; a bar is labelled by its start time
- SPY: 78 returns per full day, 43 on the 12 half-days; bars off the 5-minute grid removed; missing bars inside a day carry the last price
- Bitcoin: 288 returns per day; days with fewer than 95% of the bars dropped (88 of 747 calendar days, mostly a data gap in October 2025)

A Model for Intraday Prices

- Notation: day t has M intervals of length $\Delta = 1/M$, returns $r_{t,i}$, daily return $r_t = \sum_i r_{t,i}$
 - ▶ Variances in %²; annualised volatility $\sqrt{252 \times \text{variance}}$ (SPY), $\sqrt{365 \times \text{variance}}$ (Bitcoin); the SPY overnight return (close to next open) is outside the bars
- Log price in continuous time: $dp(s) = \mu(s) ds + \sigma(s) dW(s)$, with W a Brownian motion ([Chapter 11](#))
 - ▶ In words: over an instant ds the log price moves by a drift $\mu(s) ds$ (the instantaneous mean return) plus a random shock $\sigma(s) dW(s)$, with $dW(s) \sim N(0, ds)$
 - ▶ $\sigma(s)$ is the **spot volatility**: it changes within the day and from day to day
- **Integrated variance** of day t : the quantity we want to measure
 - ▶ $IV_t = \int_{t-1}^t \sigma^2(s) ds$: the spot variance accumulated over the day
- If σ is independent of W (no leverage) and the drift is zero (or fixed by the path of σ), then given that path the daily return is Normal: $r_t \mid \sigma \sim N(\int \mu, IV_t)$
 - ▶ With leverage (▶ [Appendix](#): Itô semimartingales) this conditional normality fails, but the limit theory of RV below still holds

Definition: Realised Variance

- **Realised variance:** the sum of squared intraday returns

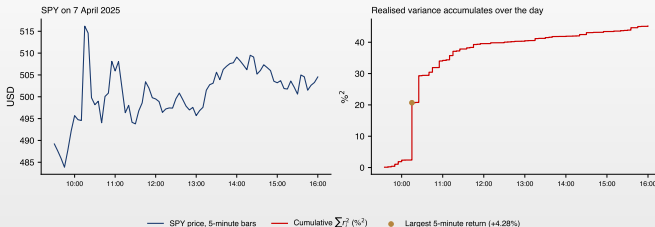
- ▶
$$RV_t = \sum_{i=1}^M r_{t,i}^2, \text{ realised volatility} = \sqrt{RV_t}$$

- As $M \rightarrow \infty$, $RV_t \rightarrow IV_t$: the drift vanishes and the sum of squares converges to the **quadratic variation** Barndorff-Nielsen & Shephard (2002)

- ▶ The mean of SPY 5-minute returns is below 1% of their standard deviation: it can be ignored

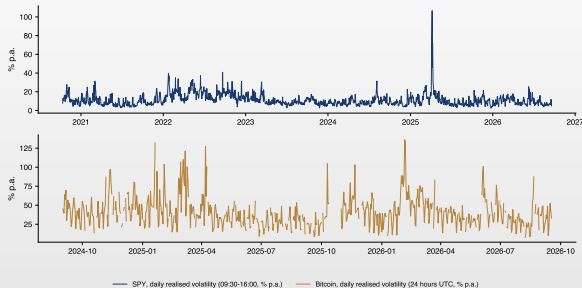
- No parameters to estimate: RV_t is a **model-free** measurement of one day's variance
- The squared daily return r_t^2 is the special case $M = 1$: unbiased (with zero drift) but extremely noisy

One Day, 78 Returns: 7 April 2025



- Left: SPY on 5-minute bars on the most volatile day of the sample; right: running sum of squared returns
- SPY opened -3.27% below the previous close, after tariff announcements; 78 five-minute returns followed
 - ▶ $RV = 45.1\%^2$: annualised realised volatility $\sqrt{252 \times 45.1} = 107\%$
- One 5-minute return of $+4.28\%$ (interval ending 10:15) contributed 41% of the day's realised variance
- Open-to-close the day ended $+3.09\%$: the daily return hides the intraday swings that RV records; by RV it is one of the most turbulent days since 2020

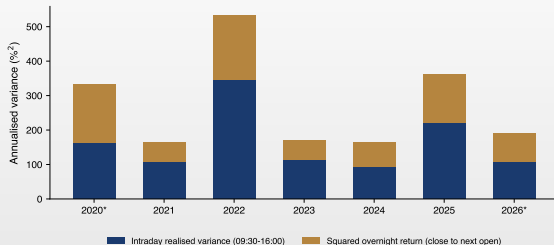
Daily Realised Volatility: SPY and Bitcoin



- Annualised, from 5-minute returns; SPY: regular US session; Bitcoin: 24 hours; the gap in October 2025 is missing data
- SPY: volatility from mean RV $\sqrt{252 \overline{RV}} = 12.9\%$ p.a., median day 9.7%, range 2.4% to 106.7%
 - ▶ Volatility computed from mean variance exceeds the median: a few very volatile days raise the root-mean-square of daily volatilities
- Bitcoin: $\sqrt{365 \overline{RV}} = 45.1\%$ p.a., median 37.8%, maximum 136.1% on 2026-02-05
- Both series show **clustering**, as with GARCH (Chapter 5), but every point is a measurement, not a model output

 MFM_ch9_rv_series

Where Is the Risk: Day or Night?



- SPY, mean daily variance by year, annualised: intraday realised variance plus squared overnight return; 2020 and 2026 are partial years
- Overnight returns carry 38% of SPY's daily variance, although the market is closed for 17.5 hours
 - ▶ News (earnings, macro data, Asian and European sessions) accumulates and is priced at the open
- Annualised: close-to-close 16.6%; open-to-close 13.6%; overnight 10.1%
- Largest overnight move: -4.08% on 2024-08-05
- For daily risk we use **total** variance: $RV_t + r_{\text{overnight},t}^2$ (annualised mean 16.4%)
 - ▶ Alternatives: scale intraday RV up, or weight the two parts optimally [Hansen & Lunde \(2005\)](#)

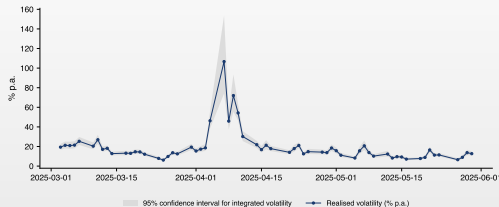
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How Precise Is Realised Variance?

- Feasible central limit theorem for RV [Barndorff-Nielsen & Shephard \(2002\)](#):

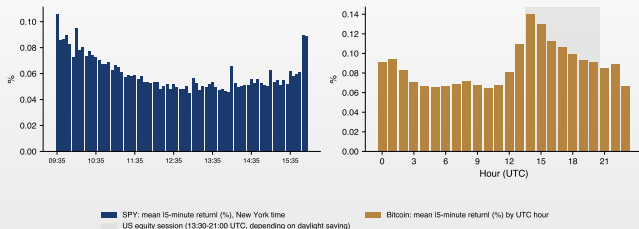
$$\sqrt{M} (RV_t - IV_t) / \sqrt{2 RQ_t} \rightarrow N(0, 1)$$
 - In words: the error $RV_t - IV_t$, scaled by $\sqrt{M}$ and by its estimated standard deviation, is approximately standard Normal
 - $IQ_t = \int_{t-1}^t \sigma^4(s) ds$: the integrated quarticity, which sets the variance of RV; it is estimated by the realised quarticity $RQ_t = \frac{M}{3} \sum_i r_{t,i}^4$
- Standard error: $se(RV_t) = \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$
- 95% CI on the log scale (always positive): $RV_t \exp(\pm 1.96 se(RV_t) / RV_t)$
 - With constant volatility, $se/RV \approx \sqrt{2/M}$: 16% for $M = 78$
- The limiting variance $2 IQ_t$ is itself random (a mixed Normal limit); replacing it by $2 RQ_t$ is legitimate because the convergence is **stable** (formal statement in the [Appendix](#))

Confidence Intervals for Daily Volatility



- SPY, March–May 2025; band: 95% confidence interval for $\sqrt{252 IV_t}$ from realised quarticity
- Median relative width of the interval (for volatility): 37% of its level
 - ▶ The band assumes a continuous price and negligible noise: on a jump day RV tends to $IV_t + \sum_j \kappa_j^2$, and $\sum r^4$ keeps the term κ^4 , so the band has no nominal coverage for either the total variation or IV_t (section Jumps)
- 9 April 2025 (the day the tariff pause was announced): volatility 72%, CI [55%, 93%]
 - ▶ The interval widens when a few returns dominate the sum: $\sum r^4$ grows faster than $(\sum r^2)^2$
- Still far more precise than r_t^2 , whose relative error is of order $\sqrt{2} \approx 141\%$ under the Normal distribution
- Practical rule: a single day's volatility is known to about $\pm 20\%$ (95% band), averages over weeks much more precisely

Intraday Seasonality

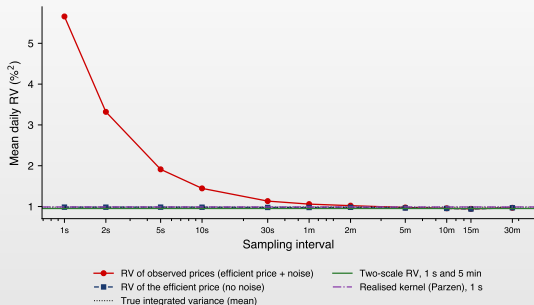


- Left: SPY, mean $|r_{t,i}|$ by 5-minute interval of the day (full days only); right: Bitcoin by UTC hour
- SPY: a **U-shape**: 0.106% in the first 5 minutes, 0.050% around midday, 0.089% in the last 5 minutes [Andersen & Bollerslev \(1997\)](#)
 - ▶ The opening auction incorporates overnight news; the close concentrates index and fund rebalancing
- Bitcoin trades 24/7 but is most volatile at 14:00 UTC, near the US equity opening (09:30 New York: 13:30 UTC in summer, 14:30 UTC in winter); quietest at 10:00 UTC (ratio 2.2)
- RV sums over the day, so seasonality does not bias it; it does bias BV and makes the jump test (section Jumps) over-reject; dividing returns by a periodicity factor f_i [Boudt, Croux & Laurent \(2011\)](#) cuts SPY rejections at the 5% level from 235 to 167 ([Appendix](#))

Why Not Use Every Tick?

- Observed prices are not the efficient price: $p_i^{\text{obs}} = p_i^* + \varepsilon_i$, so $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$
 - ▶ p_i^* : the efficient (true) log price, ε_i : the noise in the observed price, r_i^* : the efficient return
 - ▶ Sources of ε : bid-ask bounce, tick size (price discreteness), stale quotes ([Chapter 10](#))
- With i.i.d. noise of variance ω^2 and n returns per day [Zhang, Mykland & Aït-Sahalia \(2005\)](#), [Hansen & Lunde \(2006\)](#)
 - ▶ $\mathbb{E}[\text{RV}_t^{(n)} \mid \sigma] = \text{IV}_t + 2n\omega^2$ (zero drift, no leverage): each squared return carries $2\omega^2$ of noise, so the bias **grows** with the sampling frequency
 - ▶ Adjacent returns share ε_{i-1} with opposite signs: $\text{cov}(r_i, r_{i-1}) = -\omega^2$
- Trade-off: $\mathbb{E}[(\text{RV}^{(n)} - \text{IV})^2] \approx 2 \text{IQ}/n + 4n^2\omega^4$, discretisation error plus squared bias
 - ▶ MSE (mean squared error) optimal number of returns $n^* = (\text{IQ}/(4\omega^4))^{1/3}$ [Bandi & Russell \(2008\)](#): more returns cut the first term, but raise the second; derivation in the [Appendix](#)
- Numbers: $\text{IV} = 1\%^2$, $\text{IQ} = 1$, $\omega = 0.01\%$: bias 0.016 at 5 minutes, 4.680 at 1 second; $n^* = 292$ (one return every 80 s)

The Volatility Signature Plot: Simulation



- 100 simulated days, one price per second; efficient price with random daily volatility plus i.i.d. noise, $\omega = 0.01\%$
- True mean IV = 0.98; RV of observed prices: 5.66 at 1 second, 1.06 at 1 minute, 0.98 at 5 minutes
 - ▶ The bias matches $2n\omega^2$ from the previous slide
- Without noise, RV is flat in the sampling interval: the rise at high frequency is the fingerprint of noise
- Noise-robust estimators on 1-second data: two-scale RV 0.95, realised kernel 0.99
 - ▶ Error standard deviation across days: 5-minute RV 0.187, two-scale 0.150, kernel 0.045: using all data pays off only with a noise correction

 MFM_ch9_noise_simulation

Remedies: Sparse Sampling and Subsampling

- **Sparse sampling:** use one return every 5 minutes instead of every tick
 - ▶ Simple and robust; throws away most of the data [Liu, Patton & Sheppard \(2015\)](#)
- **Subsampling:** compute RV on K shifted grids (offsets $0, 1, \dots, K - 1$) and average; $RV_t^{(k)}$: RV on the grid with offset k
 - ▶ $RV_t^{\text{avg}} = \frac{1}{K} \sum_{k=0}^{K-1} RV_t^{(k)}$: same bias as one sparse grid, smaller variance
- **Two-scale RV (TSRV)** [Zhang, Mykland & Aït-Sahalia \(2005\)](#): remove the remaining bias with the all-data RV
 - ▶ $TSRV_t = RV_t^{\text{avg}} - \frac{\bar{n}}{n} RV_t^{\text{all}}$, $\bar{n} = (n - K + 1)/K$; RV_t^{all} : RV on all n returns, which estimates the noise term $2n\omega^2$; divide by $1 - \bar{n}/n$ in small samples
 - ▶ $\bar{n}$ is the average number of returns per sparse grid; our grids also keep the open and the close, so $\bar{n}$ is their actual average count

Realised Kernels

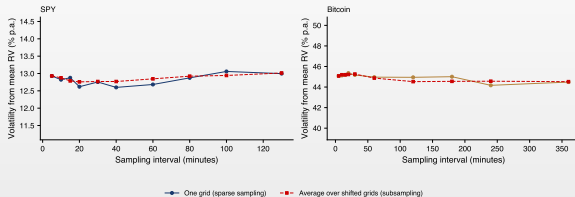


Neil Shephard, 2004

Photo: Renate Schmid (2004); CC BY-SA
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- **Realised kernel** Barndorff-Nielsen, Hansen, Lunde & Shephard (2008): add weighted autocovariances of returns
 - ▶ $RK_t = \gamma_0 + 2 \sum_{h=1}^H k\left(\frac{h}{H+1}\right) \gamma_h$, $\gamma_h = \sum_i r_{t,i} r_{t,i-h}$
 - ▶ Parzen weight function k : 1 at 0, falling smoothly to 0 at 1; keeps RK positive
- Noise makes γ_1 negative; the kernel adds it back and cancels the bias
- Bandwidth H grows with the noise-to-signal ratio: $H \approx 3.5 \xi^{4/5} n^{3/5}$, with $\xi^2 = \omega^2 / \sqrt{IQ}$ the noise-to-signal ratio; γ_h : the h -th realised autocovariance
- On liquid 5-minute data the kernel changes RV only slightly (next slide); it pays off with tick data

The Signature Plot on Real Data



- Annualised volatility from mean RV by sampling interval: one grid vs the average over all shifted grids
- Flat: SPY 12.9% at 5 minutes, 12.8% at 30, 13.0% at 130; Bitcoin 45.1% at 5 minutes, 44.5% at 6 hours
 - ▶ First-order autocorrelation of 5-minute returns: -0.008 (SPY), 0.005 (Bitcoin): no visible noise at 5 minutes
- Limit of this dataset: 5-minute bars can diagnose 5-minute and coarser sampling only; they say nothing about noise below 5 minutes
- On 5-minute SPY data: two-scale $RV / RV = 0.97$, realised kernel / $RV = 0.95$; correlations with RV 0.89 and 0.85
- Consistent with [Liu, Patton & Sheppard \(2015\)](#): almost 400 estimators on 31 assets rarely beat 5-minute RV; noise-robust estimators pay off when noise is large relative to the signal (tick or second-level data, illiquid assets)

Prices Jump: Bipower Variation



Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA

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- Add jumps to the model: $dp = \mu ds + \sigma dW + dJ$, with J a jump process (news, Chapter 11)
 - ▶ Quadratic variation now has two parts: $RV_t \rightarrow IV_t + \sum_j \kappa_j^2$, with κ_j the jump sizes in day t
 - ▶ Why separate them? Jumps are less persistent than diffusive volatility, and they price differently (options, Chapter 12)
- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns
 - ▶ $BV_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}| \rightarrow IV_t$
 - ▶ $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi} = 0.798$ for $Z \sim N(0, 1)$, so $\mu_1^{-2} = \pi/2 = 1.571$; $\frac{M}{M-1}$: finite-sample normalisation
- A jump enters one return, which is multiplied by a small neighbour: its effect on BV vanishes as M grows; the jump part of the day is $\max(RV_t - BV_t, 0)$

A Test for Jumps

- Ratio statistic [Huang & Tauchen \(2005\)](#), [Barndorff-Nielsen & Shephard \(2006\)](#):

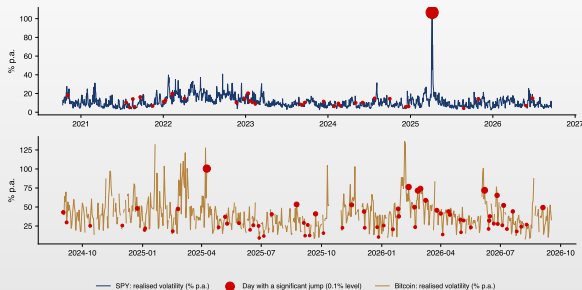
$$z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\theta M^{-1} \max(1, TQ_t/BV_t^2)}}$$

► $\theta = \pi^2/4 + \pi - 5 = 0.609$; tripower quarticity $TQ_t = M \mu_{4/3}^{-3} \frac{M}{M-2} \sum_i |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$,
 $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$

- In words: the relative jump share $(RV_t - BV_t)/RV_t$ divided by its standard error; θ : the asymptotic variance constant; TQ estimates IQ_t robustly to jumps

- Under no jumps $z_t \rightarrow N(0, 1)$; a jump makes $RV > BV$ and z_t large and positive
- One test per day: at level α we expect $\alpha \times$ days false alarms
 - We use $\alpha = 0.1\%$ (critical value 3.09) for the jump days plotted on the next slide
- Significant day: jump part $J_t = RV_t - BV_t$, continuous part $C_t = RV_t - J_t$; otherwise $J_t = 0$

Jump Days: SPY and Bitcoin



- Red points: days with a significant jump at the 0.1% level; point size proportional to the jump part J_t
- SPY: 35 jump days of 1 489 (2.4%); jumps are 2.8% of total realised variance
 - ▶ On a jump day the jump is on average 35% of that day's RV; BV/RV averages 0.95 over all days
 - ▶ Largest statistic: 2021-08-27 ($z = 5.2$), jump in the 10:00–10:05 interval, when the Federal Reserve chair's Jackson Hole speech was released
- Bitcoin: 67 jump days of 659 (10.2%), but only 1.8% of variance
 - ▶ Many small jumps on quiet days (weekends): one isolated move dominates a calm day
- The turbulent days of April 2025 are mostly **not** jump days: many large returns in a row raise BV as well

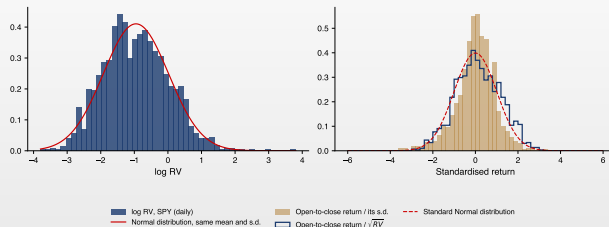
Many Days, Many Tests: Spurious Jumps

- ▣ 1 489 daily tests at 5%: 235 rejections against 74 expected under no jumps; which of them are false?
- ▣ [Benjamini & Hochberg \(1995\)](#): sort the p -values, reject the k smallest, with k the largest index such that $p_{(k)} \leq kq/N$ ($p_{(k)}$: the k -th smallest p -value, N : number of tests, q : target FDR); the FDR, $\mathbb{E}[\text{false/rejections}]$, stays $\leq q$ (with false/rejections $:= 0$ when nothing is rejected)
 - ▶ The guarantee needs valid p -values that are independent or positively dependent [Benjamini & Yekutieli \(2001\)](#); daily jump tests are only approximately so
 - ▶ SPY: FDR at 5%: 42 jump days; at 1%: 9; Bonferroni (FWER 5%): 7; FDR 5% after the periodicity correction: 17
- ▣ [Bajgrowicz, Scaillet & Treccani \(2016\)](#): without multiplicity control most detected jumps are spurious; the true ones are rare and cluster on news
- ▣ [Christensen, Oomen & Podolskij \(2014\)](#): at tick frequency most 5-minute “jumps” are steep but continuous bursts of volatility (fact or friction)
- ▣ A daily test answers “did day t contain a jump?”; locating the jump within the day needs an intraday test (Lee–Mykland, [▶ Appendix](#))

Check Yourself: Part I

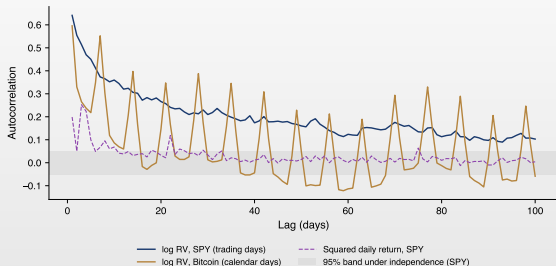
- Why does RV converge to integrated variance as M grows?
- Why does the standard error of RV use fourth powers of returns?
- Why is SPY's daily risk understated if you use intraday RV alone?
- Why does sampling every tick overstate variance?
- What does a flat signature plot tell you?
- What does a flat signature plot not tell you?
- Why does bipower variation stay close to IV on a jump day while RV does not?
- With 1 489 daily jump tests, why is " $p < 0.05$ " not evidence of a jump?
- What does the FDR control?

Distributions: log RV and Standardised Returns



- Left: daily log RV of SPY and the Normal density; right: open-to-close returns scaled by their standard deviation (filled) and by $\sqrt{\text{RV}_t}$ (outline)
- SPY RV: skewness 20.0, kurtosis 550; log RV: skewness 0.29, kurtosis 3.20 (the Normal distribution: 0 and 3)
 - ▶ Bitcoin log RV: skewness -0.27 , kurtosis 3.33: model volatility in logs
- Open-to-close returns: kurtosis 15.73; divided by $\sqrt{\text{RV}_t}$: 2.65, standard deviation 1.05
 - ▶ Bitcoin: from 7.57 to 2.77
- Heavy tails of daily returns ([Chapter 1](#)) come mostly, ex post, from changing volatility: approximately a **mixture** of Normal distributions with variances IV_t , exact only without leverage and jumps (section Realised Variance)

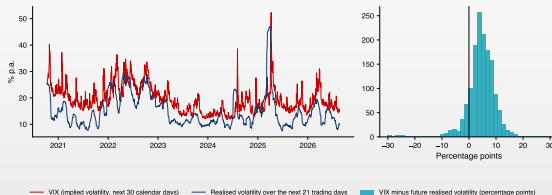
Persistence of Realised Volatility



- Autocorrelations up to 100 days: log RV of SPY (total variance) and Bitcoin, squared daily SPY returns
- SPY log RV: 0.64 at lag 1, 0.24 at one month, still 0.10 at 100 days: **highly persistent** dependence
 - ▶ An AR(1) would decay geometrically: $0.64^{22} \approx 0.00005$; a finite-sample ACF is consistent with long memory but cannot prove it
- Squared daily returns show the same persistence, hidden by noise: 0.20 at lag 1
- Bitcoin: peaks at 7, 14, 21, ... days: a **weekly cycle**
 - ▶ Weekend RV is only 0.41 of weekday RV (Saturday 27% vs Wednesday 48% p.a.): no equity or macro news, fewer institutional traders

 MFM_ch9_acf

Implied vs Realised Volatility



- Left: VIX and SPY realised volatility over the next 21 trading days; right: the gap between them
- VIX: the market's 30-day volatility implied by S&P 500 options ([Chapter 12](#)); mean 19.5% vs 15.2% realised afterwards; VIX was higher on 88% of 1468 days, by 4.4 percentage points on average
- **Variance risk premium (VRP)** [Carr & Wu \(2009\)](#), [Bollerslev, Tauchen & Zhou \(2009\)](#):

$$VRP_t = \mathbb{E}_t^Q[RV_{t,t+21}] - \mathbb{E}_t^P[RV_{t,t+21}]$$
, risk-neutral minus physical expectation of $RV_{t,t+21}$, the realised variance over the next 21 trading days; VIX^2 approximates the first term
 - The chart shows an **ex-post** spread, a noisy proxy of this ex-ante premium; it turns negative when realised volatility exceeds the earlier VIX (March–April 2025): a forecast error, not a negative premium
- As a forecast: regressing future variance on VIX^2 gives slope 0.56 (SE 0.05), $R^2 = 0.23$: VIX is informative but biased upwards

The HAR-RV Model

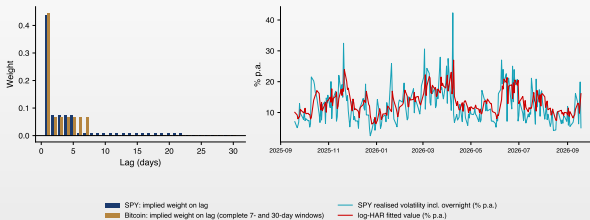
- Corsi (2009): $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$
 - ▶ $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$ (week), $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$ (month)
 - ▶ β_0 : constant; $\beta_d, \beta_w, \beta_m$: weights of yesterday's RV and of last week's and last month's average RV; ε_{t+1} : forecast error
 - ▶ Bitcoin trades every day: week = 7 days, month = 30 days
- Heterogeneous market hypothesis Müller et al. (1997): daily, weekly and monthly traders; a few components with different persistence mimic long memory
- An AR(22) with only 3 free slopes: weight $\beta_d + \beta_w/5 + \beta_m/22$ on lag 1, $\beta_w/5 + \beta_m/22$ on lags 2–5, $\beta_m/22$ on lags 6–22
- Estimation by OLS; errors are heteroskedastic and autocorrelated: use Newey–West (HAC) standard errors
 - ▶ log-HAR: the same regression for $\ln RV$; forecast $\exp(\hat{y} + \hat{s}^2/2)$, $\hat{y}$ the fitted $\ln RV$, $\hat{s}^2$ the residual variance: the conditional mean if the errors are $N(0, s^2)$ (lognormal approximation)
 - ▶ For non-Normal errors use the smearing factor $\frac{1}{T} \sum_t e^{\hat{\varepsilon}_t}$ Duan (1983); with heteroskedastic errors both are approximations

HAR Estimates: SPY and Bitcoin

	β_0	β_d	β_w	β_m	R^2
SPY, HAR (levels)	0.273 (0.064)	0.399 (0.089)	0.222 (0.096)	0.114 (0.053)	0.31
SPY, log-HAR	-0.079 (0.027)	0.362 (0.037)	0.337 (0.048)	0.167 (0.044)	0.47
Bitcoin, HAR (levels)	1.496 (0.412)	0.412 (0.089)	0.409 (0.122)	-0.065 (0.097)	0.34
Bitcoin, log-HAR	0.223 (0.094)	0.380 (0.041)	0.465 (0.080)	-0.022 (0.098)	0.36

- Full samples; SPY: total daily variance (intraday RV plus squared overnight return), 1 467 days; Bitcoin: 24-hour RV, 547 days on which all three HAR regressors are available
- Newey–West standard errors in brackets
- SPY: all three components are significant; in logs $\hat{\beta}_d = 0.362$, $\hat{\beta}_w = 0.337$, $\hat{\beta}_m = 0.167$
 - ▶ Persistence $\hat{\beta}_d + \hat{\beta}_w + \hat{\beta}_m = 0.87$: shocks fade slowly, as in GARCH ([Chapter 5](#))
- $R^2 = 0.47$ for log variance vs 0.31 for variance in levels: not comparable, the dependent variables differ; the out-of-sample losses on a common target (section Forecasting: HAR vs GARCH) decide
- Bitcoin: daily and weekly components matter; the monthly coefficient is close to 0 (-0.022 , SE 0.098)
 - ▶ Short sample (two years) and a strong weekly cycle: the week absorbs most of the memory

Implied Lag Weights and In-Sample Fit



- Left: weights of log-HAR on each past day (Bitcoin: complete 7- and 30-day windows); right: SPY, last 250 days, realised volatility and log-HAR fitted values
- SPY: yesterday gets 0.437, each of days 2–5 gets 0.075, each of days 6–22 gets 0.008
- The step-shaped weights decay slowly: the month still holds weight $22 \times 0.008 \approx 0.167$
 - ▶ Three coefficients reproduce the long memory of the autocorrelation function
- Bitcoin: yesterday 0.446, days 2–7 about 0.066 each, almost nothing beyond a week
- In-sample fitted values follow the level of volatility, not the day-to-day noise
- HAR is an AR(22) with 19 restrictions ($\phi_2 = \dots = \phi_5$, $\phi_6 = \dots = \phi_{22}$); Newey–West Wald test on SPY: $W = 18.1$, $p = 0.51$, not rejected (derivation in the [Appendix](#))

Extensions of HAR

- ▣ **HAR-CJ Andersen, Bollerslev & Diebold (2007)**: separate continuous and jump parts as regressors
 - ▶ $\ln RV_{t+1} = \beta_0 + \beta_{Cd} \ln C_t + \beta_{Cw} \ln C_t^{(w)} + \beta_{Cm} \ln C_t^{(m)}$
 - ▶ $+ \beta_{Jd} \ln(1 + J_t) + \beta_{Jw} \ln(1 + J_t^{(w)}) + \beta_{Jm} \ln(1 + J_t^{(m)}) + \varepsilon_{t+1}$
 - ▶ $C_t^{(w)}, J_t^{(w)}$ ($C_t^{(m)}, J_t^{(m)}$): arithmetic means over the last 5 (22) days, as $RV_t^{(w)}$; logs are taken after averaging
- ▣ **HARQ Bollerslev, Patton & Quaedvlieg (2016)**: measurement error in RV_t attenuates the daily slope, so put less weight on yesterday's RV when it is imprecise (high realised quarticity; derivation and SPY estimates in the ▶ Appendix)
 - ▶ β_d becomes $\beta_d + \beta_Q \sqrt{RQ_t}$ with $\beta_Q < 0$
- ▣ Weekend or holiday dummies, leverage (negative returns raise future RV), implied volatility as an extra regressor
- ▣ Multi-step: regress $\sum_{j=1}^h RV_{t+j}$ directly on the same regressors

Forecasting Design

- Target: tomorrow's daily variance; SPY: total variance, 3 January 2022 to 18 September 2026 (1 180 days); Bitcoin: 1 March 2025 onwards (422 days)
 - ▶ Every forecast uses only data available the evening before
- Five models
 - ▶ log-HAR and HAR on realised variance, re-estimated daily on an expanding window
 - ▶ GARCH(1,1) with Student- t errors on daily returns since 2000 (SPY) or 2014 (Bitcoin), re-estimated every 21 days [Bollerslev \(1986\)](#)
 - ▶ EWMA (exponentially weighted moving average) with decay factor $\lambda = 0.94$ ([Chapter 5](#)) and "yesterday's RV" (random walk)
- HAR sees intraday data; GARCH and EWMA see only daily returns

Evaluating Against a Noisy Target

- True variance is unobserved; we compare forecasts F_t with the proxy RV_t
 - ▶ A loss is **robust** if the ranking by *expected* loss with a proxy that is conditionally unbiased (given the forecasts' information) equals the ranking with true variance [Patton \(2011\)](#)
- Two robust losses
 - ▶ $MSE = (RV_t - F_t)^2$: dominated by the most volatile days
 - ▶ $QLIKE = \frac{RV_t}{F_t} - \ln \frac{RV_t}{F_t} - 1 \geq 0$: depends only on the ratio; penalises under-prediction more
- Common mistake: MAE (mean absolute error), or MSE on volatility instead of variance, is **not** robust
 - ▶ [Andersen & Bollerslev \(1998\)](#): against r_t^2 , GARCH forecasts explained a few % of the variation; against RV the same forecasts explain far more: the proxy was noisy, not the model

Tests: Diebold–Mariano and Mincer–Zarnowitz

- Diebold & Mariano (1995): loss difference $d_t = L(RV_t, F_t^A) - L(RV_t, F_t^B)$; $t = \bar{d}/\text{se}_{\text{NW}}(\bar{d}) \rightarrow N(0, 1)$
 - ▶ $L(RV_t, F_t)$: the loss (QLIKE or MSE) of forecast F_t ; $\bar{d}$: mean of d_t ; se_{NW} : Newey–West standard error
 - ▶ $t < 0$: model A has the lower loss; HAC standard error because d_t is autocorrelated (Chapter 8)
- Mincer & Zarnowitz (1969): $RV_t = a + b F_t + u_t$: regress the realised target on the forecast (a : intercept, b : slope, u_t : error); an unbiased forecast has $a = 0$, $b = 1$; $b < 1$: forecasts too extreme
 - ▶ Joint Wald test with Newey–West covariance; R^2 measures how much of the variation the forecast explains
 - ▶ R^2 is capped by the noise in RV: even the true variance would not reach 1 Andersen & Bollerslev (1998)

Results: SPY, 2022–2026

Model	QLIKE	DM vs log-HAR (p)	MSE	MZ b	MZ R^2
log-HAR	0.310	–	4.15	1.16	0.30
HAR	0.326	–1.95 (0.052)	4.48	0.75	0.26
GARCH(1,1)- t	0.343	–2.30 (0.021)	4.87	0.70	0.20
EWMA	0.393	–4.02 (< 0.001)	5.14	0.74	0.14
Yesterday's RV	0.502	–5.95 (< 0.001)	5.41	0.54	0.29

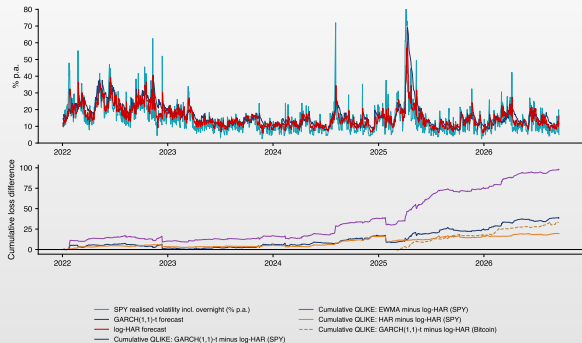
- 1 180 one-day-ahead forecasts of total daily variance; DM: $d_t = \text{QLIKE}(\text{log-HAR}) - \text{QLIKE}(\text{model})$, Newey–West; $t < 0$ favours log-HAR
- log-HAR has the lowest QLIKE and MSE: QLIKE 10% below GARCH(1,1)- t (DM $t = -2.30$, $p = 0.021$)
 - ▶ Against EWMA and yesterday's RV the gain is highly significant; against HAR in levels only marginally ($p = 0.052$)
- MSE differences are not significant: a few crisis days (April 2025) dominate the squared errors
- Mincer–Zarnowitz: log-HAR $b = 1.16$ ($p = 0.554$ for $a = 0$, $b = 1$), GARCH $b = 0.70$: no significant bias, but log-HAR explains more (R^2 0.30 vs 0.20)
- Yesterday's RV has a high R^2 but $b = 0.54$: right direction, far too extreme

Results: Bitcoin, 2025–2026

Model	QLIKE	DM vs log-HAR (p)	MSE	MZ b	MZ R^2
log-HAR	0.282	–	26.6	0.82	0.38
HAR	0.409	–1.22 (0.223)	27.6	0.89	0.34
GARCH(1,1)- t	0.360	–4.32 (< 0.001)	42.2	0.55	0.23
EWMA	0.346	–2.96 (0.003)	36.3	0.64	0.18
Yesterday's RV	0.466	–5.38 (< 0.001)	35.0	0.58	0.33

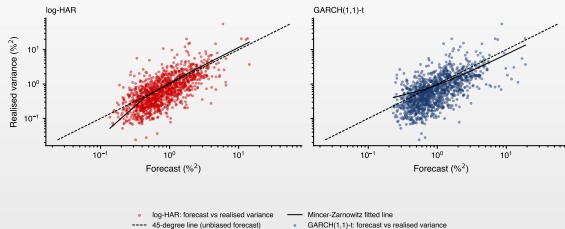
- 422 one-day-ahead forecasts of 24-hour realised variance; same models and tests as for SPY
- log-HAR wins again: QLIKE 22% below GARCH(1,1)- t (DM $t = -4.32$, $p < 0.001$)
 - ▶ GARCH on daily data is biased: MZ $b = 0.55$ ($p < 0.001$): it overreacts to large daily moves
- HAR in levels is worse than GARCH under QLIKE (0.409 vs 0.360): after calm weekends it predicts too low a variance, and QLIKE punishes under-prediction

Forecasts over Time



- Top: SPY realised volatility (with overnight) and the two forecasts, annualised; bottom: cumulative QLIKE loss of each model minus log-HAR (rising line = log-HAR better)
- After a shock both forecasts jump; GARCH stays high longer (its persistence is close to 1), log-HAR returns faster
 - ▶ April 2025: the GARCH forecast stayed above 40% p.a. until late April, while realised volatility was below 20% from mid-April
- The cumulative loss difference rises steadily, with jumps at turbulent episodes: the gain is not driven by one day
- Of the three models plotted against log-HAR, EWMA loses most: a fixed $\lambda = 0.94$ reacts too much to single days

Mincer–Zarnowitz Regressions



- SPY, 2022–2026: realised variance against the forecast (log scales); dashed: 45-degree line; black: fitted regression in levels
- log-HAR forecasts span a wider range: they separate calm from turbulent days better
- Only 1% of GARCH forecasts are below 0.26%² per day (log-HAR: minimum 0.14%²): on calm days realised variance falls far below GARCH
- The cloud is wide for both: one day's variance is hard to predict, and RV itself is measured with error
- Weekly horizon and model extensions: Seminar 9 (B6, B7)

Which Models Survive? The Model Confidence Set

Model	QLIKE	MCS p -value	In MCS (90%)
log-HAR	0.310	1.000	✓
SHAR	0.320	0.224	✓
HAR	0.326	0.014	
HARQ	0.344	0.095	
GARCH(1,1)- t	0.343	0.095	
EWMA	0.393	0.012	
Yesterday's RV	0.502	0.000	

- Hansen, Lunde & Nason (2011): test equal predictive ability of all models (range statistic, block bootstrap), drop the worst, repeat; the surviving set contains the best model with asymptotic probability $\geq 90\%$ (under the method's regularity conditions)
- SPY, 1 180 days from 2022: the 90% set is {log-HAR, SHAR}; SHAR splits yesterday's RV into positive and negative realised semivariances Patton & Sheppard (2015); estimation details in the [Appendix](#)
- A multiple-comparison procedure: many pairwise DM tests without size control would overstate the evidence ([Chapter 8](#))

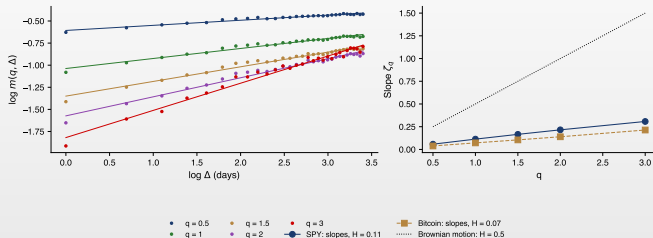
Is Volatility Rough?

- Gatheral, Jaisson & Rosenbaum (2018): log volatility behaves like a **fractional Brownian motion** with Hurst exponent $H \approx 0.1$
 - ▶ $H = 0.5$: ordinary Brownian motion; $H < 0.5$: increments are negatively correlated, paths are rougher; $H > 0.5$: smoother, trending
- Scaling law: $m(q, \Delta) = \frac{1}{T} \sum_t |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \propto \Delta^{qH}$
 - ▶ $m(q, \Delta)$: the average q -th power of the change in log volatility over Δ days; σ_t : daily volatility (here $\sqrt{RV_t}$); $\propto$: proportional to
 - ▶ Regress $\ln m(q, \Delta)$ on $\ln \Delta$: slope $\zeta_q = qH$; then H from the slopes across q
- Rough models reproduce the shape of the implied volatility smile for short maturities ([Chapter 12](#))

Rough or Noisy? The Identification Problem

- Gatheral, Jaisson & Rosenbaum (2018) report $H \approx 0.1$, but H is estimated from RV, a noisy proxy of the latent σ_t
 - ▶ Cont & Das (2024): the scaling estimator applied to realised volatility finds roughness even when spot volatility is smooth: an estimation artefact
 - ▶ Bolko, Christensen, Pakkanen & Veliev (2023) (GMM that models the RV error) and Fukasawa, Takabatake & Westphal (2022) (consistent under high-frequency asymptotics) still find $H < 0.5$
- Formal question: $\ln \text{RV}_t^{1/2} = \ln \sigma_t + e_t$, with $\ln \sigma_t$ fractional of index H and e_t the measurement error of RV
 - ▶ H_0 : $H = 1/2$ (classical diffusive volatility) against H_1 : $H < 1/2$
 - ▶ Identification problem: a small H and a large $\text{Var}(e_t)$ both flatten $m(2, \Delta)$ at short lags
- The scaling estimator on noisy RV proxies is **inconsistent** for the Hurst exponent; consistency needs high-frequency asymptotics Fukasawa, Takabatake & Westphal (2022)
 - ▶ Short-run roughness can coexist with long memory Bennedsen, Lunde & Pakkanen (2022)
- Stakes: the short-maturity skew and volatility forecasts depend on H (Chapter 12)

Scaling of Log Volatility Increments

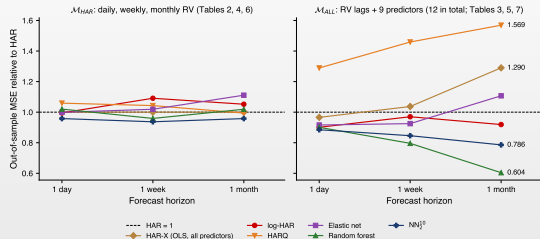


- Left: $\ln m(q, \Delta)$ against $\ln \Delta$ for SPY, $\Delta = 1, \dots, 30$ days; right: slopes ζ_q against q
- The log-log points lie close to straight lines and the slopes are close to linear in q : consistent with the scaling law, which the plot alone cannot establish
 - ▶ $\hat{H} = 0.11$ for SPY (block bootstrap 95% CI $[0.08, 0.13]$), 0.07 for Bitcoin (only observed pairs at each calendar lag; interpolating the missing days gives 0.08): far below 0.5
- Caution: RV measures volatility with error; i.i.d. error lowers $\hat{H}$, other error structures can bias it either way, so $\hat{H}$ is not a bound on the latent H
 - ▶ A simulated Brownian motion gives $\hat{H} = 0.46$; plus noise of a similar relative size, $\hat{H} = 0.18$ (Seminar 9, B8)

Case Study: Question, Data and Design

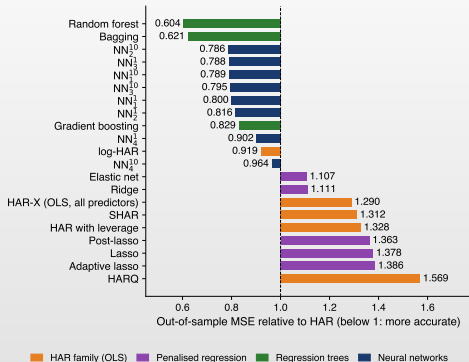
- Christensen, Siggaard & Veliyev (2023): research questions
 - ▶ does off-the-shelf machine learning (ML) forecast daily realised variance better than the HAR family?
 - ▶ if so, why?
 - ▶ HAR Corsi (2009) is linear in three RV averages, while RV is right-skewed and persistent: a nonlinear map, or predictors beyond past RV, may help
- Data: 29 of the 30 DJIA stocks, January 2001 – December 2017, $T = 4\,257$ days, 5-minute RV ($n = 78$)
 - ▶ Chronological split: training 70%, validation 10%, test 20%
- Information sets: $\mathcal{M}_{\text{HAR}}$ (daily, weekly, monthly RV) and $\mathcal{M}_{\text{ALL}}$, which adds 9 predictors: implied volatility, VIX, volume, momentum, macro and uncertainty indices (Table 1)
- Models, with minimal tuning: the HAR family by OLS, penalised regressions, tree ensembles (bagging, random forest, boosting) and neural networks; full catalogue in the [▶ Appendix](#)
- Evaluation: out-of-sample MSE relative to HAR, per stock and then averaged; one-sided DM test per stock; MCS at 90%

Case Study: Gains by Horizon



- Source: Christensen, Siggaard & Veliyev (2023), Tables 2–7: out-of-sample MSE relative to HAR, averaged over 29 DJIA stocks (test set: the last 20% of 2001–2017); left: RV lags only, right: all 12 predictors
- RV lags only: the six models shown stay between 0.937 and 1.111 times the HAR MSE at all three horizons: three RV averages leave little for a flexible model to learn
- All 12 predictors: the gain of trees and networks grows with the horizon, random forest $0.901 \rightarrow 0.796 \rightarrow 0.604$, the network with 2 hidden layers averaged over 10 fits (NN_2^{10}) $0.885 \rightarrow 0.846 \rightarrow 0.786$; the authors attribute it to the higher persistence of the ML fits (Figure 8)
- Linear models with the same predictors deteriorate (OLS $0.966 \rightarrow 1.290$, HARQ $1.289 \rightarrow 1.569$); only log-HAR stays below 1 (0.901, 0.970, 0.919)

Case Study: Ranking of 21 Models, One Month Ahead



- Source: Christensen, Siggaard & Veliyev (2023), Table 7: MSE relative to HAR, one month ahead, all 12 predictors, mean over 29 DJIA stocks; NN_k^e : k hidden layers, e networks averaged
- Tree ensembles lead: random forest 0.604, bagging 0.621; all eight networks beat HAR (0.786–0.964), but depth does not help
- All penalised regressions and linear HAR extensions lose to HAR (1.107–1.569); log-HAR (0.919) is the only linear model below 1
- Each bar averages per-stock MSE ratios: it is not a significance test of the ranking; whether it survives on SPY and under QLIKE is Seminar 9, C3

The Discovery Loop on the Roughness Question

- Question: is $\hat{H} \approx 0.1$ a property of σ_t or of the RV error e_t ? $H_0: H = 1/2$ against $H_1: H < 1/2$
- One pass of the loop with an AI assistant (reusable prompts in the [Appendix](#))
 - ▶ Literature: tabulate the published estimates of H (estimator, frequency, treatment of the RV error); check every DOI on Crossref
 - ▶ Competing hypotheses, each with its shape of $m(2, \Delta)$: fractional with $H = 0.1$; $H = 0.5$ plus i.i.d. error; two factors, one fast mean-reverting
 - ▶ Replicate the chapter's $\hat{H} = 0.11$ before extending; fix lag windows, measures and the fit with a constant noise term k in advance
- Checks the researcher owns
 - ▶ Proxy error: simulate $H = 0.5$ plus error of the same size; the procedure must not return $H \approx 0.1$
 - ▶ Overlapping increments invalidate OLS standard errors of ζ_q : use a block bootstrap; report every window tried
 - ▶ Assistants quote " $H \approx 0.1$ for all assets": check each number against the cited table

Mini-Case: Is $\hat{H}$ an Artefact of i.i.d. Error?

- A typical assistant proposal: $\hat{H} \approx 0.1$ comes from i.i.d. measurement error in RV. Implication: $m(2, \Delta) = c \Delta^{2H} + k$ with $k = 2 \text{Var}(e_t) > 0$, so $\hat{H}$ **ris**es when short lags are dropped

	$\Delta \in [1, 30]$	$[5, 30]$	$[10, 50]$	Fit with $k, [1, 50]$
Simulated $H = 0.5$ plus error	0.18	0.24	0.26	0.39
SPY, total RV	0.106	0.089	0.077	0.096
SPY, intraday RV	0.129	0.094	0.078	0.113

- First column replicates the chapter; the simulated row (one path, not a power study) shows the mechanism: $\hat{H}$ rises and the fit recovers most of the gap to 0.5
- SPY: $\hat{H}$ **falls** with the window and the fitted k is 0; $\hat{H} = 0.096$, block bootstrap 95% CI [0.058, 0.268]
- Interpretation: an illustrative diagnostic, inconsistent with the i.i.d.-error explanation; a formal rejection needs a simulation-calibrated test; the non-i.i.d. mechanism of [Cont & Das \(2024\)](#) is not tested by this design.

Your Turn: A Project Seed

- Question: does the roughness of SPY and Bitcoin volatility survive an estimator that models the RV measurement error?
- Replicate first: $\hat{H} = 0.11$ (SPY) and 0.07 (Bitcoin) from the scaling law, and $\hat{H} = 0.18$ for simulated $H = 0.5$ plus error (Seminar 9, B8)
- Deliverables
 - ▶ A pre-registration (hypotheses, estimators, lag windows, tests, simulation design) dated before the first run
 - ▶ The GMM estimator of Bolko, Christensen, Pakkanen & Veliyev (2023) on the course RV series, with Monte Carlo size and power at $H \in \{0.1, 0.5\}$
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in AI_USE.md and every error the assistant made in AI_ERRORS.md: github.com/danpele/MFM/tree/main/project

Which Tool When?

Situation	Suggested tool
Liquid asset, 1–5 minute bars	RV at 5 minutes, subsampled
Tick data, or an illiquid stock (BVB)	Pre-averaging or flat-top realised kernel ($n^{-1/4}$; Parzen $n^{-1/5}$; ▶ Appendix : rates of convergence); check the signature plot
Daily risk of an exchange-traded asset	Total variance: RV plus squared overnight return
Separate news jumps from diffusion	BV or MedRV (median realised variance, ▶ Appendix), ratio test with FDR control; Lee–Mykland for the jump time
Daily or weekly variance forecast	log-HAR (plus jumps, weekends, RQ); benchmark GARCH
Comparing forecasts	QLIKE and MSE on variance, Diebold–Mariano, Mincer–Zarnowitz, model confidence set

- Strongest result: out of sample, log-HAR beats GARCH(1,1)- t under QLIKE by 10% (SPY) and 22% (Bitcoin); MSE comparisons are inconclusive
- Assumptions: the 5-minute rule rests on a flat signature plot at 5 minutes and coarser; the jump counts on FDR control and the periodicity correction

Check Yourself, and Next: Chapter 10

Retrieval questions

- Why is R^2 of log RV not comparable with R^2 of RV in levels?
- Why is QLIKE robust to a noisy RV proxy while MSE on volatility is not?
- Why does the MCS state a set of models rather than a winner?
- How can measurement error in RV make volatility look rough?

: market microstructure

- Where does the noise ε of this chapter come from: limit order books, bid–ask spreads, tick sizes
- Liquidity, market impact and high-frequency trading, on equity and crypto exchanges

Appendix: Itô Semimartingales and Stable Convergence

- Itô semimartingale: $dp = \mu ds + \sigma dW + dJ$, with μ, σ adapted (known from the past at each time) and σ càdlàg (right-continuous with left limits, so it may jump); σ may depend on W (leverage)
 - ▶ Without jumps: $\sqrt{M}(\text{RV}_t - \text{IV}_t) \xrightarrow{\mathcal{L}-s} \text{MN}(0, 2 \text{IQ}_t)$, a **mixed** Normal distribution whose variance 2IQ_t , with integrated quarticity IQ , $\text{IQ}_t = \int \sigma^4(s) ds$, is itself random [Barndorff-Nielsen & Shephard \(2002\)](#)
- **Stable** convergence ($\mathcal{L}$ -s): the limit holds jointly with every variable of the path's σ -field [Mykland & Zhang \(2009\)](#)
 - ▶ Hence $(\text{RV}_t - \text{IV}_t) / \sqrt{2 \text{RQ}_t / M} \rightarrow N(0, 1)$: studentising by $\sqrt{2 \text{RQ}_t / M}$, with $\text{RQ}_t \rightarrow \text{IQ}_t$ random, is legitimate only under stable convergence
- Finite samples: the t -statistic for $\ln \text{RV}_t$ has better coverage than the one in levels [Barndorff-Nielsen & Shephard \(2005\)](#); the confidence intervals of the lecture use it

Appendix: Deriving the Noise Bias and the Optimal Frequency

- Observed return $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$, with ε i.i.d. $(0, \omega^2)$ and independent of p^*
 - ▶ Given the path of σ (zero drift, no leverage: σ independent of W):
 $\mathbb{E}[\text{RV}^{(n)} \mid \sigma] = \sum_i \mathbb{E}[r_i^{*2} \mid \sigma] + \sum_i \mathbb{E}[(\varepsilon_i - \varepsilon_{i-1})^2] = \text{IV} + 2n\omega^2$; $\text{cov}(r_i, r_{i-1}) = -\mathbb{E}[\varepsilon_{i-1}^2] = -\omega^2$
 - ▶ By a law of large numbers for the noise terms, $\text{RV}^{(n)}/(2n) \xrightarrow{P} \omega^2$ as $n \rightarrow \infty$: the all-tick RV estimates the noise, not IV
[Bandi & Russell \(2008\)](#)
- MSE: $\mathbb{E}[(\text{RV}^{(n)} - \text{IV})^2] \approx \underbrace{2\text{IQ}/n}_{\text{discretisation}} + \underbrace{4n^2\omega^4}_{\text{squared bias}} + O(n)$
 - ▶ First-order condition $-2\text{IQ}/n^2 + 8n\omega^4 = 0$ gives $n^* = (\text{IQ}/(4\omega^4))^{1/3}$ [Bandi & Russell \(2008\)](#)
- Numbers: $\text{IV} = 1\%^2$, $\text{IQ} = 1$, $\omega = 0.01\%$: bias 0.016 at 5 minutes, 4.680 at 1 second; $n^* = 292$ (one return every 80 s)

Appendix: Intraday Seasonality and the Jump Test

- RV sums over the day, so seasonality does not bias it; it does bias BV: $\mathbb{E}|r_i| |r_{i-1}| \propto \sigma_i \sigma_{i-1} \leq (\sigma_i^2 + \sigma_{i-1}^2)/2$
 - ▶ Fast changes of scale (open, close) push RV – BV up, so the ratio test (section Jumps) over-rejects
- Periodicity factor f_i , standard-deviation estimator of [Boudt, Croux & Laurent \(2011\)](#): scale each return by $\sqrt{\text{BV}_t/M_t}$, take the root mean square by interval of the day, normalise the mean of f_i^2 to 1
 - ▶ SPY: $f = 1.78$ at the open, 0.73 at the minimum, 1.52 at the close
- Dividing $r_{t,i}$ by f_i before the test: SPY rejections at the 5% level fall from 235 to 167; at the 0.1% level 20 remain

Appendix: Rates of Convergence under Noise

Estimator	Error of order	Positive	Source
RV on all ticks	inconsistent (bias $2n\omega^2$)	yes	Bandi & Russell (2008)
Sparse RV, fixed 5-minute grid	$M^{-1/2}$, M fixed; bias stays	yes	Liu, Patton & Sheppard (2015)
Two-scale RV	$n^{-1/6}$	no	Zhang, Mykland & Aït-Sahalia (2005)
Multi-scale RV	$n^{-1/4}$	no	Zhang (2006)
Realised kernel, flat-top / Parzen	$n^{-1/4}$ / $n^{-1/5}$	no / yes	Barndorff-Nielsen, Hansen, Lunde & Shephard (2008)
Pre-averaging	$n^{-1/4}$	no	Jacod, Li, Mykland, Podolskij & Vetter (2009)

- $n^{-1/4}$ is the optimal rate with i.i.d. noise: the price of noise is half the usual exponent
- With $n = 23\,400$ seconds: $n^{1/4} \approx 12.4$, against $\sqrt{78} \approx 8.8$ for 5-minute RV: a heuristic only (rates ignore constants, noise size and the bias of sparse RV), consistent with the modest gains in Liu, Patton & Sheppard (2015)



Appendix: Pre-Averaging

- Local averages of $k_n \approx \theta\sqrt{n}$ returns (θ : a tuning constant, not the jump-test θ) with a weight g on $[0, 1]$, e.g. $g(x) = \min(x, 1 - x)$ [Jacod, Li, Mykland, Podolskij & Vetter \(2009\)](#)
 - ▶ $\bar{r}_i = \sum_{j=1}^{k_n-1} g(j/k_n) r_{i+j}$: noise variance shrinks like $1/k_n$, the signal keeps order k_n/n
- Estimator, with $\psi_1 = \int_0^1 g'(x)^2 dx$, $\psi_2 = \int_0^1 g(x)^2 dx$ (1 and 1/12 for the triangular g ; g' : the derivative of g)
 - ▶ $\widehat{IV} = \frac{1}{\theta \psi_2 \sqrt{n}} \sum_i \bar{r}_i^2 - \frac{\psi_1}{2 \theta^2 \psi_2 n} \sum_i r_i^2$ (second term: noise bias correction)
- Rate $n^{-1/4}$; the same device gives jump-robust (pre-averaged bipower) and covariance estimators
- Built for tick data: on 5-minute bars little noise is left to average out (empirical signature plot, section Microstructure Noise)

Appendix: Realised Covariance and Asynchronous Trading

- Realised covariance $\sum_i r_{t,i}^{(1)} r_{t,i}^{(2)} \rightarrow \int \sigma_{12}(s) ds$ needs prices of both assets on the **same** grid; $r^{(1)}, r^{(2)}$: the returns of assets 1 and 2, $\sigma_{12}(s)$: their spot covariance
 - ▶ Epps effect [Epps \(1979\)](#): measured correlation falls towards 0 as sampling gets finer, because trades are asynchronous and the last price is stale
- [Hayashi & Yoshida \(2005\)](#): sum the products of **all** pairs of returns whose time intervals overlap
 - ▶ $\sum_{i,j} r_i^{(1)} r_j^{(2)} 1\{(t_{i-1}, t_i] \cap (s_{j-1}, s_j] \neq \emptyset\}$: $(t_{i-1}, t_i], (s_{j-1}, s_j]$: the intervals between consecutive trades of each asset; unbiased without synchronisation (in the absence of noise)
- With noise as well: pre-averaged or multivariate kernel versions; application: intraday portfolio risk ([Chapter 4](#))

Appendix: Locating Jumps Within the Day, Lee–Mykland

- **Lee & Mykland (2008):** $L_{t,i} = r_{t,i} / \hat{\sigma}_{t,i}$, with $\hat{\sigma}_{t,i}^2$ the bipower average of the previous $K = 270$ returns (their value for 5-minute data)
 - ▶ Under no jumps, $\max_i |L_i|$ over n returns has a Gumbel limit: reject if $(|L| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$, with $C_n = \sqrt{2 \ln n} - \frac{\ln \pi + \ln \ln n}{2\sqrt{2 \ln n}}$ and $S_n = 1/\sqrt{2 \ln n}$, the centring and scaling constants of the Gumbel limit (our $\hat{\sigma}^2$ includes the factor $\pi/2$, so the constant $c = \sqrt{2/\pi}$ of Lee & Mykland (2008) becomes 1)
 - ▶ SPY, returns divided by f_i , $\alpha = 1\%$, $n = 115\,453$ (the whole sample): threshold $|L| > 5.41$
- 169 jumps on 105 days; 62 of them on Wednesdays between 14:00 and 15:00, when FOMC statements (14:00) and press conferences (14:30) are released
 - ▶ Largest: $L = -22.9$ on 2025-10-29; only 10% of these days are also flagged by the daily ratio test
- The two tests ask different questions: one extreme return relative to local volatility, versus a daily gap between RV and BV

Appendix: Jump-Robust Estimators of Integrated Variance

- Truncated RV [Mancini \(2009\)](#): $TRV_t = \sum_i r_{t,i}^2 1\{|r_{t,i}| \leq c \hat{\sigma}_t f_i \Delta^\varpi\}$, $\varpi \in (0, 1/2)$
 - ▶ Keep only the squared returns below a threshold: c : a multiple, $\hat{\sigma}_t$: a jump-robust daily volatility, f_i : the periodicity factor, ϖ : the exponent of the interval length Δ
 - ▶ Diffusive increments are $O_p(\Delta^{1/2})$ (of order $\sqrt{\Delta}$) and end up below the threshold; jumps are $O(1)$ and are cut off
- MedRV [Andersen, Dobrev & Schaumburg \(2012\)](#): $\frac{\pi}{6-4\sqrt{3}+\pi} \frac{M}{M-2} \sum_i \text{med}(|r_{i-1}|, |r_i|, |r_{i+1}|)^2$
 - ▶ med: the median of three adjacent absolute returns; the first fraction makes MedRV consistent for IV
 - ▶ An isolated return larger than both neighbours never enters the median (which shifts to a neighbour), so a large isolated jump is excluded, though its effect is not removed exactly in finite samples; MedRV is less sensitive than BV to isolated zero returns; two adjacent jumps can still enter the median
- Jump share of SPY RV: BV 6.3%, MedRV 7.2%, TRV 12.5% ($c = 3$, $\varpi = 0.49$; truncation bites on 44% of days: c is a tuning choice)
- [Aït-Sahalia & Jacod \(2009\)](#): compare power variations $\sum |r|^p$, $p > 2$, at Δ and $k\Delta$; the ratio tends to 1 with jumps and to $k^{p/2-1}$ without: known limits under both hypotheses

Appendix: HAR as a Restricted AR(22), the Wald Test

- Unrestricted: $\ln RV_{t+1} = \phi_0 + \sum_{k=1}^{22} \phi_k \ln RV_{t+1-k} + \varepsilon_{t+1}$
 - ▶ HAR imposes $\phi_2 = \dots = \phi_5$ (3 restrictions) and $\phi_6 = \dots = \phi_{22}$ (16): $R\phi = 0$ with 19 rows
- Wald: $W = (R\hat{\phi})^\top (R\hat{V}_{\text{NW}}R^\top)^{-1}R\hat{\phi} \rightarrow \chi^2_{19}$ under the HAR restrictions; R : the 19×23 restriction matrix, $\hat{V}_{\text{NW}}$: Newey–West covariance of $\hat{\phi}$; large W = restrictions rejected
- SPY log total variance: $W = 18.1$, $p = 0.51$; R^2 0.477 (AR(22)) vs 0.470 (HAR)
 - ▶ The step restrictions are not rejected: HAR saves 19 parameters at almost no in-sample cost, and less estimation noise helps out of sample

Appendix: Measurement Error in the Regressors and HARQ

- $RV_t = IV_t + u_t$ with $\mathbb{E}[u_t | \sigma] = 0$ and $\text{Var}(u_t | \sigma) = 2IQ_t/M$ (the CLT of the section Realised Variance)
 - ▶ If $IV_{t+1} = \beta_0 + \beta IV_t + \eta_{t+1}$, OLS of RV_{t+1} on RV_t gives $\text{plim } \hat{\beta} = \beta \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}[2IQ/M]} < \beta$ for $\beta > 0$: attenuation (η_{t+1} : the shock to IV; plim: probability limit)
- If the signal variance does not rise with IQ_t (a working approximation), the signal-to-noise ratio of RV_t falls when IQ_t is high; HARQ approximates the falling weight by $\beta_d + \beta_Q \sqrt{RQ_t}$, $\beta_Q < 0$ [Bollerslev, Patton & Quaedvlieg \(2016\)](#)
- SPY, levels, Newey–West SE: intraday RV $\hat{\beta}_Q = -0.0088$ (0.0042); total variance -0.0022 (0.0024)
 - ▶ Daily slope (intraday RV) 1.13 at the median $\sqrt{RQ}$, 1.09 at its 99th percentile: the sign predicted by the theory, a small effect

Appendix: Beyond HAR, Semivariances and Joint Return–RV Models

- Realised semivariances $RS_t^\pm = \sum_i r_{t,i}^2 \mathbf{1}\{r_{t,i} \gtrless 0\}$, $RS^+ + RS^- = RV$; SHAR [Patton & Sheppard \(2015\)](#): $\beta_d RV_t$ becomes $\beta^+ RS_t^+ + \beta^- RS_t^-$
 - ▶ SPY, levels (overnight return counted as one more return of the day): $\hat{\beta}^- = 0.56$ (0.23), $\hat{\beta}^+ = 0.21$ (0.18); Wald $\beta^+ = \beta^-$: $p = 0.35$
 - ▶ Bad volatility is more persistent, as in [Patton & Sheppard \(2015\)](#), but six years of data do not separate the two slopes
 - ▶ In the MCS comparison, HARQ and SHAR are estimated in levels with the filter of [Bollerslev, Patton & Quaedvlieg \(2016\)](#): forecasts outside the [min, max] range of the estimation-sample values are replaced by the sample mean
- Realized GARCH [Hansen, Huang & Shek \(2012\)](#): a GARCH equation driven by the realised measure x_t plus a **measurement equation**
 - ▶ $\ln h_t = \omega + \beta \ln h_{t-1} + \gamma \ln x_{t-1}$, $\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$, $r_t = \sqrt{h_t} z_t$: joint quasi-maximum likelihood
 - ▶ h_t : conditional variance; x_t : realised measure (e.g. RV); z_t : standardised shock; $\tau(z_t)$: leverage function; ξ , φ : measurement-equation coefficients, u_t : its error
- HEAVY models [Shephard & Sheppard \(2010\)](#): one GARCH-type equation for the daily variance, driven by lagged RV, and one for RV itself: consistent multi-step forecasts

Appendix: Case Study, Data and Models

- Data (paper, Section 2): 29 of the 30 DJIA stocks (Visa excluded), 29 January 2001 – 31 December 2017, $T = 4\,257$ days; 5-minute RV from cleaned NYSE trade prices, $n = 78$ returns a day
 - ▶ Split: training 70% (2 964 days), validation 10% (424), test 20% (847); robustness: training 1 000 or 2 000 days, validation 200 (paper, Appendix A.1)
 - ▶ $\mathcal{M}_{\text{HAR}}$: daily, weekly, monthly RV; $\mathcal{M}_{\text{ALL}}$ adds 9 predictors (Table 1): implied volatility, earnings-day dummy, 1-week momentum, change in log dollar volume; VIX, squared Hang Seng return, business-conditions index, change in the 3-month T-bill rate, policy-uncertainty index
- Models (paper, Section 1): HAR, log-HAR, HAR with leverage, SHAR, HARQ by OLS on a rolling window; ridge (L_2 penalty), lasso (L_1), elastic net (a mix), adaptive and post-lasso, penalties tuned on the validation set
 - ▶ Trees: bagging, random forest (RF) with Breiman's defaults, 500 trees, $J/3$ candidate predictors per split (J : number of predictors), minimum node size 5 (paper, Table A.6), gradient boosting
 - ▶ Neural networks NN_k^e : $k = 1, \dots, 4$ hidden layers (2; 4-2; 8-4-2; 16-8-4-2 neurons), leaky ReLU, Adam, dropout, early stopping; $e = 1$: best of 100 seeds on validation, $e = 10$: average of the best 10
- Evaluation (paper, Section 1.6): out-of-sample MSE relative to HAR, averaged over stocks; one-sided DM test per stock [Diebold & Mariano \(1995\)](#); MCS at 90% [Hansen, Lunde & Nason \(2011\)](#)

Appendix: Case Study, Complete Results

- One day ahead, RV lags only (Table 2): MSE relative to HAR: log-HAR 0.995, HARQ 1.059, RF 1.020, neural networks 0.954–0.990
 - ▶ Past RV leaves little to learn: the networks give the clearest gains (1–5%), log-HAR barely beats HAR
- One day ahead, all 12 predictors (Table 3): OLS with all predictors 0.966, HARQ 1.289; elastic net 0.916, RF 0.901, log-HAR 0.901, NN_2^{10} 0.885
 - ▶ Extra predictors pay most with shrinkage, nonlinearity or the log transform; plain OLS gains less (0.966), and HARQ overfits
- One month ahead, all predictors (Table 7): RF 0.604, bagging 0.621, networks 0.786–0.964, log-HAR 0.919, OLS with all predictors 1.290
 - ▶ The gain grows with the horizon: the ML fits are more persistent (Figure 8) and approximate the long memory of RV better
- MCS at 90% (Figure 4): with all predictors, the networks stay in the set for about 80% of stocks, HAR for about 20–40%
 - ▶ Variable importance from accumulated local effects (Figure 7): daily RV, weekly RV and implied volatility dominate in all four models shown; RF and networks rank implied volatility higher than OLS with all predictors and elastic net

Appendix: Case Study, Replication, Limits and Extensions

- Replicate on course data: SPY 5-minute bars, open-to-close RV with 78 returns a day (the sampling of the paper, Section 2); the authors report that SPY confirms the stock results (footnote 12)
 - ▶ Split from Appendix A.1 of the paper: 1 000 training days, 200 validation days, the rest to 18 September 2026 as test set
 - ▶ Predictors: VIX (computed from S&P 500 index options; a proxy for SPY's own implied volatility, which would need SPY options), squared Hang Seng return, SPY momentum and dollar volume, T-bill and policy uncertainty (FRED); no earnings dummy for a fund (Seminar 9, C3)
- Limits stated by the authors
 - ▶ Minimal tuning: more tuning could improve ML further (paper, Section 6); the networks are trained once and not re-estimated in the test window (paper, Section 2)
 - ▶ Rankings depend on the split: with 1 000 training days the financial crisis falls in the test set and the HAR extensions beat HAR (paper, Section 3.1 and Appendix A.1)
 - ▶ HAR models with a separate jump component are not included (paper, Section 1.2); accumulated local effects describe the average prediction and miss rare events such as earnings days (paper, Section 3.2)
- Open extension: the paper ranks variance forecasts by MSE only
 - ▶ does the ML advantage survive QLIKE, which is robust to the noisy RV proxy [Patton \(2011\)](#)?
 - ▶ does it hold for Bitcoin's 24-hour RV?

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: tabulate the evidence, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Tabulate the estimates of H published since 2018: estimator, sampling frequency, treatment of the RV measurement error, point estimate and standard error, DOI.
- 2. **Competing hypotheses**, each with its implied shape of $m(2, \Delta)$
 - ▶ Derive $m(2, \Delta)$ for three processes that give H near 0.1 with the scaling estimator: fractional log-volatility with $H = 0.1$; $H = 0.5$ plus i.i.d. error; a two-factor model with one fast mean-reverting factor.
- 3. **Data and code**: replicate the chapter's number first
 - ▶ Write Python that builds daily total variance from the SPY 5-minute bars of the course data (5-minute RV plus the squared overnight return, previous close to today's open), computes $m(q, \Delta)$ for q in $\{0.5, 1, 1.5, 2, 3\}$, $\Delta = 1..30$, and the slope H . Target: $\hat{H} = 0.11$

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- 4. **Robustness**, pre-registered, with a power check
 - ▶ Write a pre-registered plan: lag windows [1,30], [5,30], [10,50]; RV, bipower variation and 30-minute subsampled RV; a fit $m(2, \Delta) = c \Delta^{2H} + k$; block-bootstrap intervals; size and power by simulation at $H = 0.1$ and 0.5 .
- 5. **Critique**: ask for the strongest counter-argument
 - ▶ Act as a referee: which non-i.i.d. error structures in log RV (serially correlated, level-dependent, driven by overnight returns or jumps) would survive the constant-k correction, and how would you detect each?
- 6. **Report**: all windows and measures, bootstrap intervals, the simulation design, code and the AI log

Appendix: What the Researcher Must Check

- **Proxy error:** RV is an estimate with a known standard error; test: simulate $H = 0.5$ plus error of the same size and check that the procedure does not return $H \approx 0.1$
- **Overlapping increments:** $m(q, \Delta)$ reuses the same days at every Δ , so OLS standard errors of ζ_q are invalid; test: block bootstrap
- **Microstructure noise and jumps:** check the signature plot, then repeat with BV and with subsampled RV
- **Specification search:** lag windows and q grids chosen after seeing $\hat{H}$ inflate significance; fix them in advance and report all
- **Invented estimates:** assistants quote " $H \approx 0.1$ for all assets"; check each number against the table of the cited paper

Appendix: Further Reading

□ Textbook and surveys

- ▶ Franke, Härdle & Hafner (2019): statistics of financial markets, volatility models
- ▶ Andersen, Bollerslev, Diebold & Labys (2003): modelling and forecasting realised volatility; Corsi (2009): the HAR model

□ Crypto-assets

- ▶ Intraday information can enter Bitcoin risk forecasts; the entropy of high-frequency returns has been proposed as one such signal Pele & Mazurencu-Marinescu-Pele (2019)
- ▶ Crypto-assets behave like a distinct asset class with their own volatility dynamics Pele et al. (2023)

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Nonparametric and Semiparametric Models
- ☐  Kalman Filter
- ☐  Applied Time Series Analysis with Python

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

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